Chapter 7 · Semester 1

ROTATION OF RIGID BODIES

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21 worked examples 1 simulation Exercises with answers

7.1 Centre of mass and centre of gravity

Centre of mass (CM)

of a system is the **point at which the entire mass of the system may be considered to be concentrated**.

Centre of gravity (CG)

of a body is the **point at which the entire weight of the body may be considered to act**. The resultant of the gravitational forces on all the particles of the body passes through this point.

Centre of mass of a system of particles in a plane

The centre of mass is the mass-weighted average position of the particles:

$$\bar{x} = \frac{\Sigma m_i x_i}{\Sigma m_i} \qquad \bar{y} = \frac{\Sigma m_i y_i}{\Sigma m_i}$$

m_i = mass of the i-th particle at (x_i, y_i)

Additional example (syllabus 7.1(b)) · centre of mass of particles

Masses of 1 kg at (0, 0), 2 kg at (3 m, 0) and 2 kg at (0, 4 m) lie in a plane. Find the centre of mass.

Solution

$$\bar{x} = \frac{(1)(0) + (2)(3) + (2)(0)}{1 + 2 + 2} = \frac{6}{5} = \mathbf{1.2\ m}$$

$$\bar{y} = \frac{(1)(0) + (2)(0) + (2)(4)}{5} = \frac{8}{5} = \mathbf{1.6\ m}$$

Centre of mass at **(1.2 m, 1.6 m)**

Path of the centre of mass of a two-particle system

- Consider two particles m_1 and m_2 moving under their mutual interaction only (an explosion, or a collision), with no external force on the system.
- The internal forces are equal and opposite (Newton's third law), so they cancel and do not affect the motion of the centre of mass.
- The centre of mass therefore **keeps moving in a straight line at constant velocity** (or stays at rest), however complicated the paths of the individual particles are.
- If an external force acts, the centre of mass moves as if it were a single particle of mass ($m_1 + m_2$) acted on by that force. Example: after a shell explodes in flight, the centre of mass of the fragments continues along the original parabola.

When is the centre of mass also the centre of gravity?

- The centre of mass depends only on how the mass is distributed; the centre of gravity depends on how the weight is distributed.
- They **coincide when the gravitational field strength g is uniform** over the whole body (same magnitude and direction at every particle).
- This holds for everyday objects, which are tiny compared with the Earth. They would differ only for a very large body (a mountain, a very tall tower) over which g varies noticeably.

7.2 Moment of inertia

Inertia (linear motion) opposes a change in motion.
Moment of inertia (rotational motion) opposes a change in rotation.

> **Moment of inertia, I**
>
> of a body about an axis is the **sum of the products of the mass of each particle and the square of its perpendicular distance from the axis**. It measures the body's resistance to a change in its rotational motion.

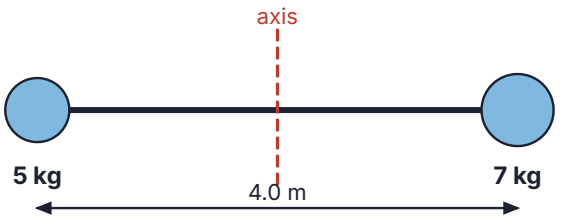
- It depends on (i) the mass, (ii) how the mass is distributed about the axis, and (iii) the position and orientation of the axis.
- It is a **scalar**; unit kg m^2.

Particle: $I = mr^2$ System of particles: $I = \Sigma m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + \ldots + m_n r_n^2$

r = perpendicular distance of each particle from the axis of rotation

Example 7.2.1

Particles of 5 kg and 7 kg are mounted 4.0 m apart on a light rod. Calculate the moment of inertia of the system (a) about an axis halfway between the masses, (b) about an axis 0.50 m to the left of the 5 kg mass.



Solution

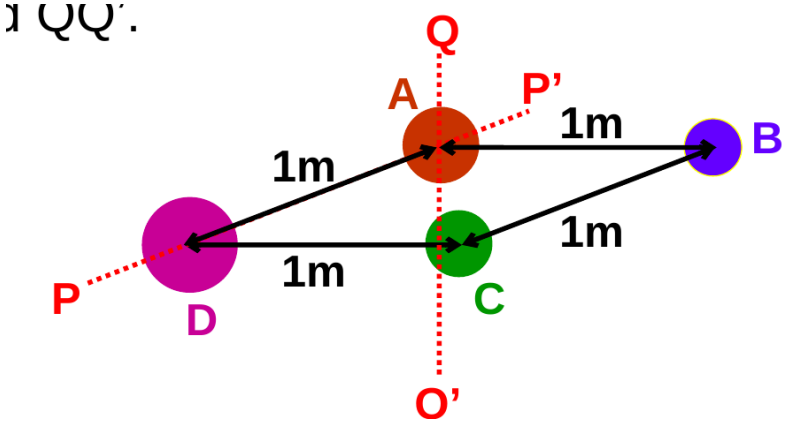
(a) $I = m_1r_1^2 + m_2r_2^2 = (5)(2)^2 + (7)(2)^2 =$ **48 kg m²**

(b) Axis 0.50 m to the left of the 5 kg mass: $I = (5)(0.50)^2 + (7)(4.50)^2 =$ **143 kg m²**

The slide gives 87 kg m² for (b). That is the answer for an axis 0.50 m to the **right** of the 5 kg mass (between the masses): $(5)(0.50)^2 + (7)(3.50)^2 = 87$ kg m².

Example 7.2.2

Calculate the moment of inertia of the system about the axis PP′ and about the axis QQ′. (m_A = 2 kg, m_B = 1 kg, m_C = 1.5 kg, m_D = 3 kg)



Solution

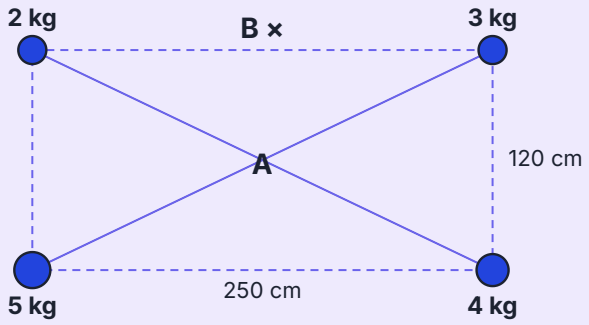
Masses on an axis contribute nothing (r = 0).

$I_{PP'} = m_B r_B^2 + m_C r_C^2 = (1)(1)^2 + (1.5)(1)^2 =$ **2.5 kg m²**

$I_{QQ'} = m_B r_B^2 + m_C r_C^2 + m_D r_D^2 =$ **7 kg m²**

Exercise: moment of inertia

1. Masses of 2 kg, 3 kg, 4 kg and 5 kg are at the corners of a 250 cm × 120 cm rectangle. Calculate their moment of inertia about an axis perpendicular to the page through (a) the centre A, (b) the midpoint B of the top edge.



Answer

(a) $d = \sqrt{1.25^2 + 0.60^2} = 1.39$ m for all four masses: $I_A = 14 \times 1.92 = 27$ kg m^2
(b) top masses $d = 1.25$ m; bottom masses $d = \sqrt{1.25^2 + 1.2^2} = 1.73$ m: $I_B = (2 + 3)(1.25)^2 + (5 + 4)(1.73)^2 = 34.8$ kg m^2 (the printed answer is 33 kg m^2)

Moments of inertia of some uniform bodies

STPM: the moment of inertia of these shapes is given in the question; you do not need to derive them.

Body	Axis	Moment of inertia
particle	distance r from the axis	mr^2
thin ring, hoop or cylindrical shell	through the centre, perpendicular to the plane	mr^2
thin rod, length L	through the centre, perpendicular to the rod	$\frac{1}{12}mL^2$
solid cylinder or disc	along the cylinder axis	$\frac{1}{2}mr^2$
solid sphere	through the centre	$\frac{2}{5}mr^2$
thin spherical shell	through the centre	$\frac{2}{3}mr^2$

Example 7.2.3

Calculate the moment of inertia of a 10.8 kg solid sphere of radius 0.648 m about an axis through its centre.

Solution

$$I = \frac{2}{5}mr^2 = \frac{2}{5}(10.8)(0.648)^2 = \mathbf{1.81\ kg\ m^2}$$

7.3 Torque and angular acceleration

Torque, τ

is the **tendency of a force to rotate a body about an axis**: the quantitative measure of the ability of a force to cause or change rotational motion.

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \quad \tau = rF\sin\theta$$

- A **vector**; unit N m.
- Sign convention: clockwise negative, anticlockwise positive.
- $r\sin\theta$ is the perpendicular distance from the axis to the line of action of **F**.

Angular acceleration, α

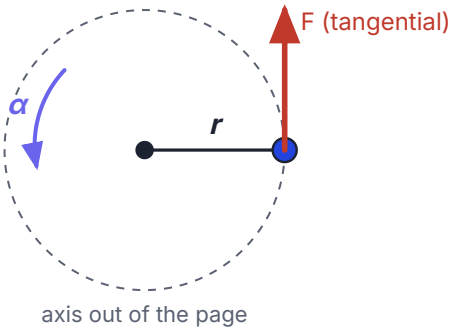
is the **rate of change of angular velocity**: $\alpha = \frac{d\omega}{dt}$ (rad s^{-2}). For a point at radius r, the tangential acceleration is $a_t = r\alpha$.

A net torque produces an angular acceleration: $\tau = I\alpha$

A constant tangential force F acts on a particle of mass m at radius r: $F = ma_t$ and $a_t = r\alpha$, so $F = mr\alpha$

Torque of this force ($\theta = 90°$): $\tau = rF = mr^2\alpha = I\alpha$ ($I = mr^2$ for a particle)

A rigid body is made of n particles: $\tau = m_1r_1^2\alpha + m_2r_2^2\alpha + \ldots = (\Sigma m_i r_i^2)\alpha$



$$\tau_{net} = I\alpha$$

the rotational form of Newton's second law, $F = ma$; τ_{net} is the net torque on the body

Example 7.3.1

A force of 200 N acts tangentially on the rim of a wheel of radius 25 cm. (a) Calculate the torque. (b) Repeat if the force makes an angle of 40° with the spoke of the wheel.

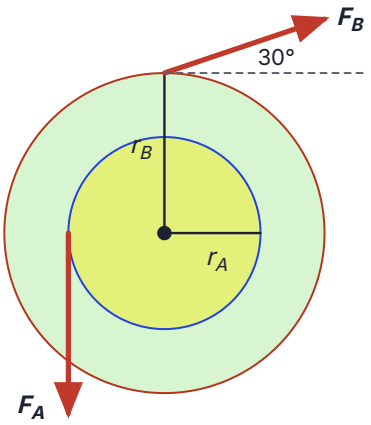
Solution

(a) $\tau = rF \sin\theta = (0.25)(200 \sin 90°) =$ **50 N m**

(b) $\tau = (0.25)(200 \sin 40°) =$ **32 N m**

Example 7.3.2

Two thin discs of radius r_A = 30 cm and r_B = 50 cm are fixed together on an axle through their centres. Calculate the net torque due to the two forces shown, each of 50 N.



Solution

$\tau_{net} = r_A F_A \sin 90° - r_B F_B \sin 60°$

$= (0.30)(50) - (0.50)(50) \sin 60° =$ **−6.7 N m** (6.7 N m clockwise)

Example 7.3.3

An 8.0 kg wheel has a radius of 30 cm. (a) What is its moment of inertia? (b) What torque gives it an angular acceleration of 3.0 rad s^{-2}? (Treat it as a uniform disc.)

Solution

(a) $I = \frac{1}{2}mr^2 = \frac{1}{2}(8.0)(0.30)^2 =$ **0.36 kg m²**

(b) $\tau = I\alpha = (0.36)(3.0) =$ **1.08 N m**

Example 7.3.4

A grinding disc of radius 0.6 m and mass 90 kg rotates at 460 rev min^{-1}. What frictional force, applied tangentially at the edge, stops it in 20 s?

Solution

$I = \frac{1}{2}mr^2 = \frac{1}{2}(90)(0.6)^2 = 16.2$ kg m^2

$\omega_0 = \frac{460 \times 2\pi}{60} = 48.2$ rad s^{-1}; $\alpha = \frac{0 - 48.2}{20} = -2.41$ rad s^{-2}

$rF = I\alpha \Rightarrow F = \frac{I\alpha}{r} = \frac{16.2(-2.41)}{0.6} = -65.1$ N: a force of **65.1 N** opposing the rotation

Example 7.3.5

What tangential force applied to the rim of a 2 kg disc-shaped wheel of radius 0.5 m gives it an angular acceleration of 4.8 rad s^{-2}? (Neglect friction.)

Solution

$I = \frac{1}{2}mr^2 = \frac{1}{2}(2)(0.5)^2 = 0.25$ kg m^2

$rF = I\alpha \Rightarrow F = \frac{0.25(4.8)}{0.5} = \mathbf{2.4\ N}$

Example 7.3.6

A 15.0 N force is applied to a cord wrapped round a pulley of mass 4.00 kg and radius 33.0 cm. There is a friction torque of 1.10 N m at the axle. Calculate the angular acceleration of the pulley.

Solution

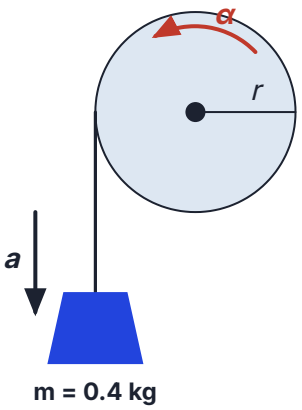
$I = \frac{1}{2}mR_0^2 = \frac{1}{2}(4.00)(0.33)^2 = 0.22$ kg m^2

$\tau_{net} = I\alpha$: $rF - \tau_{fr} = I\alpha$

$\alpha = \frac{(0.33)(15.0) - 1.10}{0.22} = \mathbf{17.5\ rad\ s^{-2}}$

Example 7.3.7

A 400 g mass hangs from a cord wound on a wheel of radius 15 cm. When released from rest it falls 2.0 m in 2 s. Calculate the moment of inertia of the wheel.



Solution

Mass (linear): $s = ut + \frac{1}{2}at^2 \Rightarrow 2.0 = \frac{1}{2}a(2)^2 \Rightarrow a = 1\ \text{m s}^{-2}$

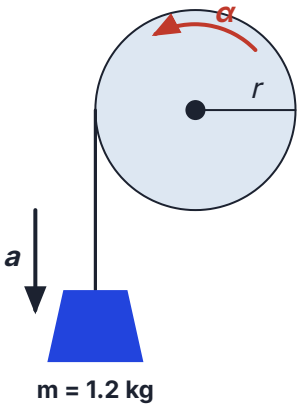
$mg - T = ma \Rightarrow T = 0.4(9.81 - 1) = 3.52\ \text{N}$

Wheel (rotation): $Tr = I\alpha$ with $\alpha = \frac{a}{r} = \frac{1}{0.15} = 6.67\ \text{rad s}^{-2}$

$I = \frac{Tr}{\alpha} = \frac{(3.52)(0.15)}{6.67} = \mathbf{0.079\ kg\ m^2}$

Example 7.3.8

A uniform disc of mass 2.5 kg and radius 20 cm is mounted on a fixed horizontal axle. A 1.2 kg block hangs from a light cord wrapped round its rim. Calculate (a) the acceleration of the block, (b) the angular acceleration of the disc, (c) the tension in the cord.



Solution

Disc (rotation): $TR = I\alpha = (\tfrac{1}{2}MR^2)(a/R) \Rightarrow T = \tfrac{1}{2}Ma$ (1)

Block (translation): $mg - T = ma$ (2)

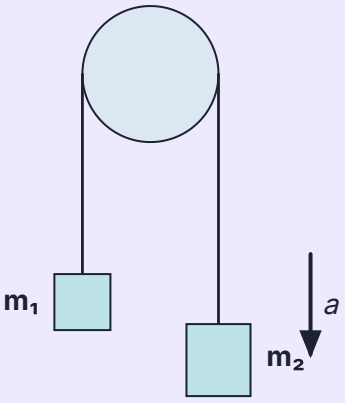
(1) into (2): $a = \dfrac{mg}{m + \tfrac{1}{2}M} = \dfrac{(1.2)(9.81)}{1.2 + 1.25} = \mathbf{4.80\ m\ s^{-2}}$

(b) $\alpha = \dfrac{a}{R} = \dfrac{4.80}{0.20} = \mathbf{24.0\ rad\ s^{-2}}$

(c) $T = \tfrac{1}{2}Ma = \tfrac{1}{2}(2.5)(4.80) = \mathbf{6.0\ N}$

Exercise: pulley with friction

1. Masses m_1 = 0.40 kg and m_2 = 0.80 kg hang from a pulley of mass 0.20 kg and radius 0.15 m. A constant friction torque of 0.35 N m acts at the axle. What is the acceleration of the masses?



Answer

$m_2g - T_2 = m_2a;\ T_1 - m_1g = m_1a;\ (T_2 - T_1)R - \tau_{fr} = \tfrac{1}{2}MR^2(a/R)$

$$a = \frac{(m_2 - m_1)g - \tau_{fr}/R}{m_1 + m_2 + \tfrac{1}{2}M} = \frac{3.92 - 2.33}{1.30} = 1.2\ \text{m s}^{-2}$$

Rotational motion with constant angular acceleration

If ω increases uniformly from ω_0 to ω in time t: $\omega = \omega_0 + \alpha t$ (1)

Angular displacement = average angular velocity × time: $\theta = \tfrac{1}{2}(\omega_0 + \omega)t$ (2)

(1) into (2): $\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$ (3)

From (1), $t = (\omega - \omega_0)/\alpha$; substitute into (2): $\omega^2 = \omega_0^2 + 2\alpha\theta$ (4)

Linear motion	Rotational motion
$v = u + at$	$\omega = \omega_0 + \alpha t$
$s = \tfrac{1}{2}(u + v)t$	$\theta = \tfrac{1}{2}(\omega_0 + \omega)t$
$s = ut + \tfrac{1}{2}at^2$	$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$
$v^2 = u^2 + 2as$	$\omega^2 = \omega_0^2 + 2\alpha\theta$

Example 7.3.9

A wheel rotates with constant angular acceleration 2.5 rad s^{-2}. At t = 0 its angular velocity is 3.0 rad s^{-1}. Calculate (a) the angle turned in the first 2 s, (b) the angular velocity at t = 2 s.

Solution

(a) $\theta = \omega_0 t + \frac{1}{2}\alpha t^2 = (3.0)(2) + \frac{1}{2}(2.5)(2)^2 =$ **11 rad**

(b) $\omega = \omega_0 + \alpha t = 3.0 + (2.5)(2) =$ **8.0 rad s^{-1}**

Example 7.3.10

A car wheel speeds up from rest to 100 rad s^{-1} in 9 s. Calculate (a) the angular acceleration, (b) the angular speed at t = 10 s, (c) the angle turned in 20 s (the acceleration continues).

Solution

(a) $\alpha = \frac{100}{9} =$ **11.11 rad s^{-2}**

(b) $\omega = (11.11)(10) =$ **111.1 rad s^{-1}**

(c) $\theta = \frac{1}{2}(11.11)(20)^2 =$ **2222 rad**

Example 7.3.11

A wheel turns with constant angular acceleration 0.45 rad s^{-2}. (a) How long does it take to reach 8.00 rad s^{-1} from rest? (b) Through how many revolutions does it turn in this time?

Solution

(a) $t = \frac{\omega - \omega_0}{\alpha} = \frac{8.00}{0.45} =$ **17.8 s**

(b) $\theta = \frac{\omega^2}{2\alpha} = \frac{64}{0.90} = 71.1$ rad $= \frac{71.1}{2\pi} =$ **11.3 rev**

Exercises: uniform angular acceleration

1. The angular speed of a disc decreases uniformly from 12.00 to 4.00 rad s^{-1} in 16.0 s. Calculate the angular acceleration and the number of revolutions made in this time.

   Answer

   $\alpha = -0.500$ rad s^{-2}; $\theta = \frac{1}{2}(12 + 4)(16) = 128$ rad = 20.4 rev

2. A flywheel of radius 0.08 m speeds up from 5 rpm to 11 rpm in 100 s. Calculate (a) its angular acceleration, (b) the tangential acceleration of a point on the rim.

   Answer

   (a) 6.3×10^{-3} rad s^{-2} (b) $a_t = \alpha r = 5.04 \times 10^{-4}$ m s^{-2}

7.4 Angular momentum and its conservation

Angular momentum, L

of a particle about an axis is the **cross product of its position vector r and its linear momentum p**: **L = r × p**.

Magnitude: $L = rp \sin\theta$. For circular motion, **r** ⊥ **v** ($\theta = 90°$) and $p = mv$, so $L = mvr$

With $v = r\omega$: $L = mr^2\omega$

A rigid body is made of many particles: $L = (\Sigma m_i r_i^2)\omega = I\omega$

$$L = I\omega$$

vector (right-hand rule, along the axis); unit kg m^2 s^{-1}

Example 7.4.1

A 10 kg disc of radius 0.25 m has an angular momentum of 0.45 kg m^2 s^{-1}. What is its angular speed?

Solution

$$I = \tfrac{1}{2}(10)(0.25)^2 = 0.3125 \text{ kg m}^2; \quad \omega = \frac{L}{I} = \frac{0.45}{0.3125} = \mathbf{1.44 \text{ rad s}^{-1}}$$

Example 7.4.2

A 500 g steel rod 30 cm long is pivoted about its centre and rotates at 300 rev min^{-1}. What is its angular momentum?

Solution

$$I = \frac{1}{12}mL^2 = \frac{1}{12}(0.5)(0.3)^2 = 3.75 \times 10^{-3}\ \text{kg m}^2;\ \omega = \frac{300 \times 2\pi}{60} = 31.4\ \text{rad s}^{-1};\ L = I\omega = \mathbf{0.118\ kg\ m^2\ s^{-1}}$$

Principle of conservation of angular momentum

If **no external torque** acts on a system, its **total angular momentum is conserved** (constant).

$$I_1\omega_1 = I_2\omega_2$$

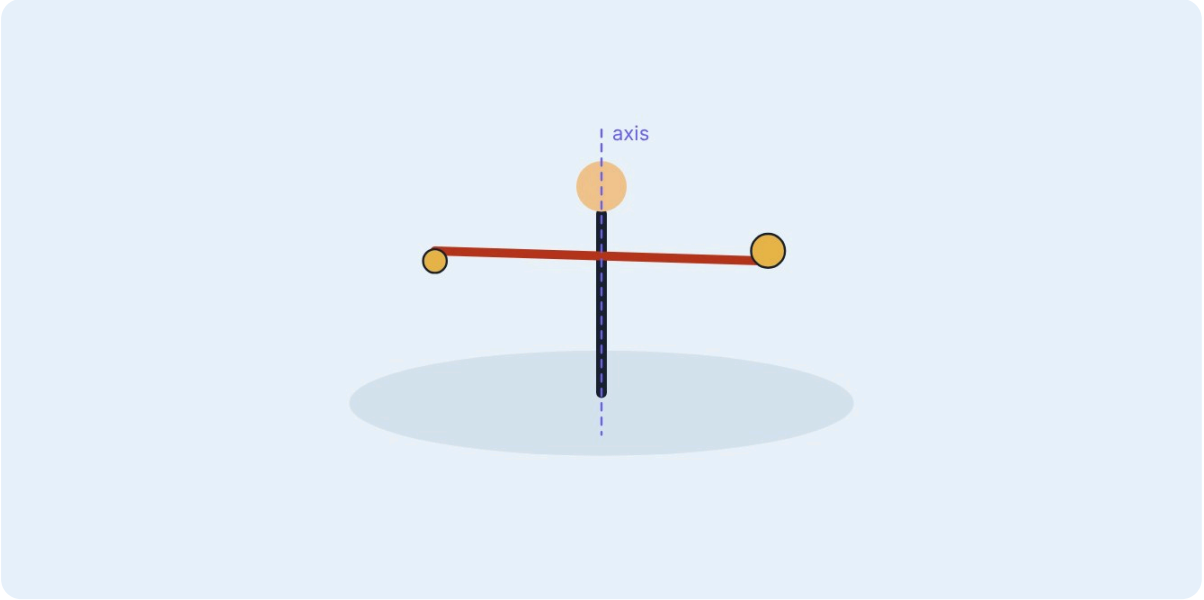
An ice skater starts spinning with her arms stretched out. Bringing her arms close to the axis reduces her moment of inertia I; since $I\omega$ is constant, her angular velocity ω increases. Divers, acrobats and gymnasts use the same idea.



Simulation

Conservation of angular momentum: the spinning skater

Pull the arms in and out. L = Iω stays the same, so when I gets smaller ω gets bigger. The kinetic energy ½Iω² is not conserved: the skater does work pulling her arms in.



moment of inertia I	angular velocity ω = L/I	L = Iω (constant)	rotational KE ½Iω²
5.86 kg m²	**1.02 rad/s**	**6.00 kg m²/s**	**3.1 J**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 7 to use the simulation.

Example 7.4.3

A horizontal disc rotates at 100 rpm about a vertical axis through its centre. A 10 g particle is dropped onto the disc 9.0 cm from the axis and sticks. The angular velocity falls to 90 rpm. Find the moment of inertia of the disc.

Solution

No external torque: $I\omega_0 = (I + mr^2)\omega$

$mr^2 = (0.010)(0.090)^2 = 8.1 \times 10^{-5}$ kg m^2

$I(100) = (I + 8.1 \times 10^{-5})(90) \Rightarrow$ **$I = 7.3 \times 10^{-4}$ kg m^2**

Example 7.4.4

A flywheel (moment of inertia 95 kg m^2) rotates at 60 rad s^{-1}. A coaxial gear wheel (5.0 kg m^2) is initially at rest. The two are joined together. Calculate the common angular velocity and the heat generated.

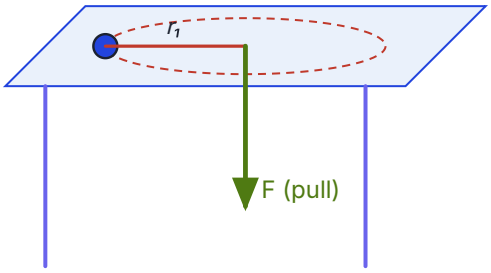
Solution

(a) $I_1\omega_{01} + I_2\omega_{02} = (I_1 + I_2)\omega \Rightarrow (95)(60) + 0 = (100)\omega \Rightarrow$ **$\omega = 57$ rad s^{-1}**

(b) Heat = loss in rotational kinetic energy = $\frac{1}{2}(95)(60)^2 - \frac{1}{2}(100)(57)^2 = 171\,000 - 162\,450 =$ **8550 J**

Example 7.4.5

A mass on a string moves in a circle on a frictionless table; the string passes through a hole in the table. Initially the speed is 2.4 m s^{-1} at radius 0.80 m. The string is pulled slowly until the radius is 0.48 m. What is the new speed?



Solution

The pull acts through the axis, so there is no torque: $L_{\text{initial}} = L_{\text{final}}$

$mr_1^2\omega_1 = mr_2^2\omega_2$ and $v = r\omega \Rightarrow v_1r_1 = v_2r_2$

$$v_2 = \frac{(2.4)(0.80)}{0.48} = \textbf{4.0 m s}^{-1}$$

Exercise: conservation of angular momentum

1. A merry-go-round of diameter 4.5 m rotates freely at 0.70 rad s^{-1}; its moment of inertia is 1750 kg m^2. Four people of 65 kg each step onto its edge. What is the new angular velocity?

Answer

$\Delta I = 4(65)(2.25)^2 = 1316$ kg m^2; $I_2 = 3066$ kg m^2; $\omega_2 = \dfrac{(1750)(0.70)}{3066} = \mathbf{0.40\ rad\ s^{-1}}$

7.5 Rotational kinetic energy

Rotational kinetic energy

is the **kinetic energy of a body due to its rotation**: the sum of the kinetic energies of all its particles, for rotation about a fixed axis (no translation).

$K_R = \tfrac{1}{2}m_1v_1^2 + \tfrac{1}{2}m_2v_2^2 + \dots + \tfrac{1}{2}m_nv_n^2$, with $v = r\omega$

$= \tfrac{1}{2}\omega^2(m_1r_1^2 + m_2r_2^2 + \dots) = \tfrac{1}{2}(\Sigma m_ir_i^2)\omega^2$

$$K = \tfrac{1}{2}I\omega^2$$

Example 7.5.1

A wheel (uniform disc) of mass 6.0 kg and radius 40 cm rotates at 300 rpm. Calculate its moment of inertia and its rotational kinetic energy.

Solution

$\omega = \dfrac{300 \times 2\pi}{60} = 31.4$ rad s^{-1}

$I = \tfrac{1}{2}mr^2 = \tfrac{1}{2}(6.0)(0.40)^2 = \mathbf{0.48\ kg\ m^2}$

$K = \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}(0.48)(31.4)^2 = \mathbf{2.37 \times 10^2\ J}$

Linear and rotational quantities compared

Linear motion	Rotational motion
displacement, s	angular displacement, θ
velocity, v	angular velocity, ω
acceleration, a	angular acceleration, α
mass, m	moment of inertia, I
force, F	torque, τ
$F = ma$	$\tau = I\alpha$
momentum, $p = mv$	angular momentum, $L = I\omega$
work, $W = Fs$	work, $W = \tau\theta$
$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$

Relationships: $s = r\theta$, $v = r\omega$, $a = r\alpha$; $\omega_{ave} = \theta/t = (\omega + \omega_0)/2$.